

Christian DiPietrantonio

ME 5130: Advanced Heat and Mass Transfer

Homework 01: Heat Conduction

September 15, 2026

1–2, Problem Statement: Verify that ∇T and $\nabla \cdot \mathbf{q}$ in the spherical coordinate system (r, ϕ, θ) are given as:

$$\nabla T = \hat{\mathbf{u}}_r \frac{\partial T}{\partial r} + \hat{\mathbf{u}}_\phi \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} + \hat{\mathbf{u}}_\theta \frac{1}{r} \frac{\partial T}{\partial \theta} \quad (1)$$

$$\nabla \cdot \mathbf{q} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 q_r) + \frac{1}{r \sin \theta} \frac{\partial q_\phi}{\partial \phi} + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (q_\theta \sin \theta) \quad (2)$$

1–2, Solution: The transformation from spherical to cartesian coordinates is:

$$\begin{aligned} x &= r \sin \theta \cos \phi \\ y &= r \sin \theta \sin \phi \\ z &= r \cos \theta \end{aligned} \quad (3)$$

and the following can be defined:

$$\begin{aligned} u_1 &= r \\ u_2 &= \phi \\ u_3 &= \theta \end{aligned} \quad (4)$$

Now, from Ö zısıık, we have Eq. (1-19):

$$\nabla T = \hat{\mathbf{u}}_1 \frac{1}{\alpha_1} \frac{\partial T}{\partial u_1} + \hat{\mathbf{u}}_2 \frac{1}{\alpha_2} \frac{\partial T}{\partial u_2} + \hat{\mathbf{u}}_3 \frac{1}{\alpha_3} \frac{\partial T}{\partial u_3} \quad (5)$$

Eq. (1-22a):

Eq. (1-22b):

$$\alpha = \alpha_1 \alpha_2 \alpha_3 \quad (6)$$

and Eq (1-18):

1–12, Problem Statement: Set up the mathematical formulation of the following heat conduction problems:

1. A slab in $0 \leq x \leq L$ is initially at a temperature $F(x)$. For times $t > 0$, the boundary at $x = 0$ is kept insulated and the boundary at $x = L$ dissipates heat by convection into a medium at zero temperature.
2. A semiinfinite region $0 \leq x < \infty$ is initially at a temperature $F(x)$. For times $t > 0$, heat is generated in the medium at a constant rate of $g_0 W/m^3$, while the boundary at $x = 0$ is kept at zero temperature.

3. A solid cylinder $0 \leq r \leq b$ is initially at a temperature $F(r)$. For times $t > 0$, heat is generated in the medium at a rate of $g(r)$, W/m^3 , while the boundary at $r = b$ dissipates heat by convection into a medium at zero temperature.
4. A solid sphere $0 \leq r \leq b$ is initially at temperature $F(r)$. For times $t > 0$, heat is generated in the medium at a rate of $g(r)$, W/m^3 , while the boundary at $r = b$ is kept at a uniform temperature T_0 .

1–12, Solution:

1–15, Problem Statement: A long cylindrical iron bar of diameter $D = 5$ cm, initially at temperature $T_0 = 650^\circ\text{C}$, is exposed to an air stream at $T_\infty = 50^\circ\text{C}$. The heat transfer coefficient between the air stream and the surface of the bar is $h = 80 \text{ W m}^{-2}^\circ\text{C}^{-1}$. Thermophysical properties may be taken as $\rho = 7800 \text{ kg m}^{-3}$, $C_p = 460 \text{ J kg}^{-1}^\circ\text{C}^{-1}$, and $k = 60 \text{ W m}^{-1}^\circ\text{C}^{-1}$. Determine the time required for the temperature of the bar to reach 250°C by using the lumped system analysis.

1–15, Solution:

1–16, Problem Statement: A thermocouple is to be used to measure the temperature in a gas stream. The junction may be approximated as a sphere having thermal conductivity $k = 25 \text{ W m}^{-1}^\circ\text{C}^{-1}$, $\rho = 8400 \text{ kg m}^{-3}$, and $C_p = 0.4 \text{ kJ kg}^{-1}^\circ\text{C}^{-1}$. The heat transfer coefficient between the junction and the gas stream is $h = 560 \text{ W m}^{-2}^\circ\text{C}^{-1}$. Calculate the diameter of the junction if the thermocouple should measure 95% of the applied temperature difference in 3s.

1–16, Solution:

2–2, Problem Statement: A slab, $0 \leq x \leq L$, is initially at a temperature $F(x)$. For times $t > 0$ the boundary surface at $x = 0$ is kept insulated and that at $x = L$ dissipates heat by convection into a medium at zero temperature with a heat transfer coefficient h . Obtain an expression for the temperature distribution $T(x, t)$ in the slab for times $t > 0$ and for the heat flux at the boundary surface $x = L$.

2–2, Solution:

2–6, Problem Statement: In a one-dimensional infinite medium, $-\infty < x < \infty$, initially, the region $a < x < b$ is at a constant temperature T_0 , and everywhere outside this region is at zero temperature. Obtain an expression for the temperature distribution $T(x, t)$ in the medium for times $t > 0$.

2–6, Solution:

2–15, Problem Statement: Obtain an expression for the steady-state temperature distribution $T(x, y)$ in a rectangular region $0 \leq x \leq a$, $0 \leq y \leq b$ for the following boundary conditions: the boundary at $x = 0$ is kept insulated, the boundary at $y = 0$ is kept at temperature $f(x)$ and the boundaries at $x = a$ and $y = b$ dissipate heat by convection into an environment at zero temperature. Assume the heat transfer coefficient to be the same for both boundaries.

2–15, Solution: